



Republic of the Philippines
OFFICE OF THE PRESIDENT
COMMISSION ON HIGHER EDUCATION



CHED MEMORANDUM ORDER

No. 87
Series of 2017

**SUBJECT: POLICIES, STANDARDS AND GUIDELINES FOR THE BACHELOR
OF SCIENCE IN COMPUTER ENGINEERING (BSCpE) EFFECTIVE
(AY) 2018-2019**

In accordance with the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "*Higher Education Act of 1994*," in pursuance of an outcomes-based quality assurance system as advocated under CMO 46 s. 2012 (Policy-Standard to Enhance Quality Assurance (QA) in Philippine Higher Education through an Outcomes-Based and Typology-Based Quality Assurance) and as addendum to CMO 37, s. 2012 (Establishment of an Outcomes-Based Educational System in Higher Education Institutions offering Engineering Programs), and by virtue of Commission en banc Resolution No. 788-2017 dated October 24, 2017 the following Policies, Standards and Guidelines (PSG) are hereby adopted and promulgated by the Commission.

**ARTICLE I
INTRODUCTION**

Section 1. Rationale

Based on the *Guidelines for the Implementation of CMO No. 46 series of 2012* and CMO 37 s. 2012, this PSG implements shift to outcomes based education leading to competency based standards. It specifies the "core competencies" expected of BS Computer Engineering graduates "regardless of the type of Higher Education Institutions (HEI) they graduate from." However, in recognition of outcomes-based education (OBE) and the typology of HEIs, this PSG also provide ample space for HEIs to innovate in the curriculum in line with the assessment of how best to achieve learning outcomes in their particular contexts and their respective missions.

**ARTICLE II
AUTHORITY TO OPERATE**

Section 2. Government Recognition

All private higher education institutions (PHEIs) intending to offer BS Computer Engineering must first secure proper authority from the Commission in accordance with this PSG. All PHEIs with an existing BS Computer Engineering program are required to shift to an outcomes-based approach based on CMO 37, s. 2012 and guided by this PSG. State universities and colleges (SUCs), and local universities and

colleges (LUCs) should likewise strictly adhere to the provisions in these policies and standards.

ARTICLE III GENERAL PROVISIONS

Per Section 13 of RA 7722, the higher education institution shall exercise academic freedom in its curricular offerings but must comply with the minimum requirements for specific academic programs, the general education distribution requirements and the specific professional courses.

Section 3. Minimum Standards

The Articles that follow give minimum standards and other requirements and guidelines. The minimum standards are expressed as a minimum set of desired program outcomes which are given in Article IV Section 6. CHED designed a curriculum to attain such outcomes. This curriculum is shown in Article V Section 10 and Section 11 as **sample curriculum**. The number of units of this curriculum is here prescribed as the “minimum unit requirement” under Section 13 of RA 7722. To assure alignment of the curriculum with the program outcomes, this PSG provides a sample curriculum map in Article V Section 12 for the HEI to refer to in compliance with the implementing guidelines of CMO 37, s.2012.

Using a learner-centered/outcomes-based approach, CHED provided a description of Outcomes-Based Teaching and Learning delivery method in Article V Section 13. A sample course syllabus is also given in Article V Section 14 as support to the outcomes-based delivery method. Based on the curriculum and the means of its delivery, CHED determines the physical resource requirements for the library, laboratories and other facilities and the human resource requirements in terms of Administration and faculty. These are provided for in Article VI.

Section 4. Curriculum Design

The HEIs are allowed to design curricula suited to their own contexts and missions provided that they can demonstrate that the same leads to the attainment of the required minimum set of outcomes, albeit by a different route. In the same vein, they have latitude in terms of curriculum delivery and in terms of specification and deployment of human and physical resources as long as they can show that the attainment of the program outcomes and satisfaction of program educational objectives can be assured by the alternative means they propose.

The HEIs can use the **CHED Implementation Handbook for Outcomes-Based Education (OBE) and the Institutional Sustainability Assessment (ISA)** as a guide in making their submissions for Sections 19 to 24 of Article VII.



ARTICLE IV PROGRAM SPECIFICATIONS

Section 5. Program Description

5.1 Degree Name

The degree program described herein shall be called Bachelor of Science in Computer Engineering (BSCpE).

5.2 Nature of the Field of Study

The Bachelor of Science in Computer Engineering (BSCpE) is a program that embodies the science and technology of design, development, implementation, maintenance and integration of software and hardware components in modern computing systems and computer-controlled equipment.

5.3 Characteristics of Computer Engineering Graduates

With the ubiquity of computers, computer-based systems and networks in the world today, computer engineers must be versatile in the knowledge drawn from standard topics in computer science and electrical engineering as well as the foundations in mathematics and sciences. Because of the rapid pace of change in the computing field, computer engineers must be life-long learners to maintain their knowledge and skills within their chosen discipline.

An important distinction should be made between computer engineers, electrical engineers, other computer professionals, and engineering technologists. While such distinctions are sometimes ambiguous, computer engineers generally should satisfy the following three characteristics.

1. Possess the ability to design computers, computer-based systems and networks that include both hardware and software and their integration to solve novel engineering problems, subject to trade-offs involving a set of competing goals and constraints. In this context, "design" refers to a level of ability beyond "assembling" or "configuring" systems.
2. Have a breadth of knowledge in mathematics and engineering sciences, associated with the broader scope of engineering and beyond that narrowly required for the field.
3. Acquire and maintain a preparation for professional practice in engineering.



5.4 Program Educational Objectives

Program Educational Objectives (PEOs) are broad statements that describe the career and professional accomplishments that the program is preparing graduates to achieve within 3–5 years from graduation. PEOs are based on the needs of the program's constituencies and these shall be determined, articulated, and disseminated to the general public by the unit or department of the HEI offering the BSCpE program. The PEOs should also be assessed and evaluated periodically for continuing improvement.

5.5 Knowledge Areas

The knowledge areas include the following but not limited to:

- a) Circuits and Electronics
- b) Computing Algorithms
- c) Computer Architecture and Organization
- d) Digital Design
- e) Embedded Systems
- f) Computer Networks
- g) Professional Practice
- h) Information Security
- i) Signal Processing
- j) Systems and Project Engineering
- k) Software Design
- l) Occupational Health and Safety
- m) Technopreneurship

5.6 Allied Programs

The allied programs of the BSCpE program are the following:

- a) Electrical Engineering
- b) Electronics Engineering
- c) Software Engineering
- d) Computer Science
- e) Information Technology

These programs are those that may be considered as equivalent to the program for the purpose of determining faculty qualifications to handle allied and related courses to the program.

Section 6. Institutional and Program Outcomes

The minimum standards for the BS Computer Engineering program are expressed in the following minimum set of institutional and BSCpE program outcomes.

6.1 Institutional outcomes

- a) Graduates of professional institutions must demonstrate a service orientation in one's profession,



- b) Graduates of colleges must participate in various types of employment, development activities, and public discourses, particularly in response to the needs of the communities one serves
- c) Graduates of universities must participate in the generation of new knowledge or in research and development projects
- d) Graduates of State Universities and Colleges must, in addition, have the competencies to support “national, regional and local development plans.” (RA 7722).
- e) Graduates of higher educational institutions must preserve and promote the Filipino historical and cultural heritage.

6.2. BSCpE Program Outcomes

By the time of graduation, the students of the program shall have the ability to:

- a) Ability to apply knowledge of mathematics and science to solve complex engineering problems;
- b) Ability to design and conduct experiments, as well as to analyze and interpret data;
- c) Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards;
- d) Ability to function on multidisciplinary teams;
- e) Ability to identify, formulate, and solve complex engineering problems;
- f) Understanding of professional and ethical responsibility;
- g) Ability to communicate effectively;
- h) Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context;
- i) Recognition of the need for, and an ability to engage in life-long learning
- j) Knowledge of contemporary issues;
- k) Ability to use techniques, skills, and modern engineering tools necessary for engineering practice; and
- l) Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments.

A PHEI, SUC, or LUC, at its option, may adopt mission-related program outcomes that are not included in the minimum set

Annex I presents the Competency Standards, Attributes and competencies of a Computer Engineer which should result from the program outcomes stated above.



Section 7. Sample Performance Indicators

Performance Indicators (PIs) are specific, measurable statements identifying the performance(s) required to meet the outcome; confirmable through evidence.

Table 1. Sample Performance Indicators of a Program Outcome

Performance Outcomes		Performance Indicators	
f	Understanding of professional and ethical responsibility	1	Demonstrate knowledge of professional code of ethics
		2	Evaluate the ethical and societal implications of a design solution to a problem in CpE

Section 8. Program Assessment and Evaluation

Program Assessment refers to one or more processes that identify, collect, and prepare data to evaluate the attainment of Program Outcomes and Program Educational Objectives.

Program Evaluation pertains to one or more processes for interpreting the data and evidence accumulated from the assessment. Evaluation determines the extent at which the Program Outcomes and the Program Educational Objectives are achieved by comparing actual achievement versus set targets and standards. Evaluation results in decisions and actions regarding the continuous improvement of the program.

All HEIs are encouraged to form a Consultative Body to be part of the assessment and evaluation processes to be represented by the stakeholders.

8.1 Assessments and Evaluation of PEOs

The Assessment of Program Educational Objectives may include the following: the stakeholders of the program have to be contacted through surveys or focus group discussion to obtain feedback data on the extent of the achievement of the PEOs.

8.2. Assessment and Evaluation of POs

In the case of Program Outcomes Assessment, the defined Performance Indicators shall be connected to Key Courses (usually the Demonstrating or “D” courses in the Curriculum map), and an appropriate Assessment Methods (AM) may be applied. These methods may be direct or indirect depending on whether the demonstration of learning was measured by actual observation and authentic work of the student or through gathered opinions from the student or his peers. Refer to Table 2.



Table 2. Sample Matrix Linking Performance Indicators with Key Courses, Assessment Methods, Set Targets and Standards

Performance Indicators		Key Courses	Assessment Tools	Targets and Standards
1	Demonstrate knowledge of professional code of ethics	OJT	Employer Assessment Form (EAF)	60% of students enrolled in the course shall get at least a rating of 70%
2	Evaluate the ethical and societal implications of a design solution to a problem in CpE	Design Project 2 (Project Implementation)	Rubric for Design Presentation (RDP)	60% of students enrolled in the course shall get at least a rating of 70%

Other Methods of Program Assessment and Evaluation may be found in the *CHED Implementation Handbook for Outcomes-Based Education (OBE) and Institutional Sustainability Assessment (ISA)*.

Section 9. Continuous Quality Improvement

There must be a documented process for the assessment and evaluation of program educational objectives and program outcomes.

The comparison of achieved performance indicators with declared targets or standards of performance should serve as basis for the priority projects or programs for improving the weak performance indicators. Such projects and programs shall be documented as well as the results of its implementation. This regular cycle of documentation of projects, programs for remediation and their successful implementation shall serve as the evidence for Continuous Quality Improvement.

ARTICLE V CURRICULUM

Section 10. Curriculum Description

The BSCpE curriculum is designed to meet the SOs/POs stated in Article IV Section 6. This is articulated in a curriculum map discussed in Section 12 to develop graduates of the program to have a strong background in mathematics,



natural, physical and allied sciences. Also, it contains complementary courses such as general education courses to ensure that the graduates are articulate and understands the nature of their role and impact of their work in the society and environment.

The BSCpE curriculum is designed to guarantee breadth of knowledge of the discipline through a set of professional courses and to ensure depth and focus in certain disciplines through cognates/tracks. Also, it develops student's ability to use modern tools necessary to solve problems in the field of computer engineering.

The curriculum has a minimum total of 166 credit units, comprising of 115 units of technical courses. These technical courses include 12 units of mathematics, 8 units of natural/physical sciences, 6 units of basic engineering sciences, 8 units of allied courses, 72 units of professional courses, and 9 units of elective/cognate courses.

The general education courses in accordance with CMO 20 s. 2013 - The New General Education Curriculum consists of 24 units of general education courses, 12 units of GEC electives/mandated courses, 8 units of Physical Education (PE), and 6 units of National Service Training Program (NSTP).

Section 11. Sample Curriculum

11.1. Components:

Below is a sample curriculum of the BSCpE program. The institution may enrich the sample curriculum depending on the needs of the industry and community, provided that all prescribed courses are offered and pre-requisite and co-requisite are observed.

Classification/Field/Course	Minimum no. of hours / week		Minimum Credit Units
	Lecture	L/F/D	
I. TECHNICAL COURSES			
A. Mathematics			
Calculus 1	3	0	3
Calculus 2	3	0	3
Engineering Data Analysis	3	0	3
Differential Equations	3	0	3
Subtotal	12	0	12
B. Natural/Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers	3	3	4
Subtotal	6	6	8
C. Basic Engineering Sciences			
Computer-Aided Drafting	0	3	1
Engineering Economics	3	0	3
Technopreneurship101	3	0	3
Subtotal	6	3	7



Classification/Field/Course	Minimum no. of hours / week		Minimum Credit Units
	Lecture	L/F/D	
D. Allied Courses			
Fundamentals of Electrical Circuits	3	3	4
Fundamentals of Electronic Circuits	3	3	4
Subtotal	6	6	8
E. Professional Courses			
Discrete Mathematics	3	0	3
Numerical Methods	3	0	3
Computer Engineering as a Discipline	1	0	1
Fundamentals of Mixed Signals and Sensors	3	0	3
Computer Engineering Drafting and Design	0	3	1
Programming Logic and Design	0	6	2
Data Structures and Algorithms	0	6	2
Object Oriented Programming	0	6	2
Software Design	3	3	4
Microprocessors	3	3	4
Logic Circuits and Design	3	3	4
Methods of Research	2	0	2
Operating Systems	3	0	3
Computer Architecture and Organization	3	3	4
Data and Digital Communications	3	0	3
Computer Networks and Security	3	3	4
Embedded Systems	3	3	4
Digital Signal Processing	3	3	4
Feedback and Control Systems	3	0	3
Introduction to HDL	0	3	1
Seminars and Fieldtrips	0	3	1
Basic Occupational Health and Safety	3	0	3
CpE Laws and Professional Practice	2	0	2
Emerging Technologies in CpE	3	0	3
CpE Practice and Design 1	0	3	1
CpE Practice and Design 2	0	6	2
On the Job Training	3	240	3
Subtotal	53	297	72
F. Cognates/Electives (Please refer to Suggested Electives)			
Cognate/Track Course 1			3
Cognate/Track Course 2			3
Cognate/Track Course 3			3
Subtotal			9
II. NON - TECHNICAL COURSES			
A. General Education Courses			
Science, Technology, and Society	3	0	3
The Contemporary World	3	0	3

Classification/Field/Course	Minimum no. of hours / week		Minimum Credit Units
	Lecture	L/F/D	
Readings in Philippine History	3	0	3
Understanding the Self	3	0	3
Art Appreciation	3	0	3
Purposive Communication	3	0	3
Mathematics for the Modern World	3	0	3
Ethics	3	0	3
Subtotal	24	0	24
B. GEC Electives/Mandated Courses			
GEC Elective 1	3	0	3
GEC Elective 2	3	0	3
GEC Elective 3	3	0	3
Life and Works of Rizal	3	0	3
Subtotal	12	0	12
C. Physical Education			
PE 1	2	0	2
PE 2	2	0	2
PE 3	2	0	2
PE 4	2	0	2
Subtotal	8	0	8
D. National Service Training Program			
NSTP 1	3	0	3
NSTP 2	3	0	3
Subtotal	6	0	6
GRAND TOTAL	133	312	166

SUMMARY OF THE BScPE CURRICULUM

Classification/Field/Course	Total No. of Hours / Week		Minimum Credit Units
	Lecture	Lab	
I. TECHNICAL COURSES			
A. Mathematics	12	0	12
B. Natural/Physical Sciences	6	6	8
C. Basic Engineering Sciences	6	3	7
D. Allied Courses	6	6	8
E. Professional Courses	53	297	72
F. Cognates/Electives			9
Subtotal	83	312	116
II. NON-TECHNICAL COURSES			
A. General Education Courses	24	0	24
B. GEC Electives/Mandated Courses	12	0	12
C. Physical Education	8	0	8
D. NSTP	6	0	6
Subtotal	50	0	50
GRAND TOTAL (including PE and NSTP)	133	312	166



11.2. Program of Study

The institution may enrich the sample/model program of study depending on the needs of the industry, provided that all prescribed courses required in the curriculum outlines are offered and pre-requisites and co-requisites are complied with.

The sample Program of Study listed below is meant for HEIs operating on a Semestral System. HEIs with CHED approved trimester or quarter term systems may adjust their courses and course specifications accordingly to fit their delivery system, as long as the minimum requirements are still satisfied.

The HEIs are also encouraged to include other courses to fulfill their institutional outcomes, as long as the total units for the whole program shall not be less than **166 units**, including P.E., and NSTP.

SAMPLE SEMESTRAL PROGRAM OF STUDY

FIRST YEAR

1st year – 1st semester

Courses	No. of Hours		Units	Prerequisites
	Lec	Lab/Field/Drafting		
Calculus 1	3	0	3	
Chemistry for Engineers	3	3	4	
Computer Engineering as a Discipline	1	0	1	
Programming Logic and Design	0	6	2	
Mathematics for the Modern World	3	0	3	
Science, Technology, and Society	3	0	3	
Understanding the Self	3	0	3	
Physical Education 1	2	0	2	
NSTP 1	3	0	3	
TOTAL	21	9	24	



1st year – 2nd semester

Courses	No. of Hours		Units	Prerequisites
	Lec	Lab/Field/Drafting		
Calculus 2	3	0	3	Calculus 1
Physics for Engineers	3	3	4	Calculus 1
Object Oriented Programming	0	6	2	Programming Logic and Design
Engineering Data Analysis	3	0	3	Calculus 1
Discrete Mathematics	3	0	3	Calculus 1
Readings in Philippine History	3	0	3	
Physical Education 2	2	0	2	Physical Education 1
NSTP 2	3	0	3	NSTP 1
TOTAL	20	9	23	

SECOND YEAR

2nd year – 1st Semester

Courses	No. of Hours		Units	Prerequisites
	Lec	Lab/Field/Drafting		
Differential Equations	3	0	3	Calculus 2
Art Appreciation	3	0	3	
Data Structures and Algorithms	0	6	2	Object Oriented Programming
Engineering Economics	3	0	3	2 nd Year Standing*
Fundamentals of Electrical Circuits	3	3	4	Physics for Engineers
GEC Elective 1	3	0	3	
Computer-Aided Drafting	0	3	1	2 nd Year Standing*
Physical Education 3	2	0	2	Physical Education 2
TOTAL	17	12	21	



2nd year – 2nd semester

Courses	No. of Hours		Units	Prerequisites
	Lec	Lab/Field/Drafting		
Numerical Methods	3	0	3	Differential Equations
Software Design	3	3	4	Data Structures and Algorithms
Purposive Communication	3	0	3	
Fundamentals of Electronic Circuits	3	3	4	Fundamentals of Electrical Circuits
Life and Works of Rizal	3	0	3	
Physical Education 4	2	0	2	
The Contemporary World	3	0	3	
TOTAL	20	6	22	

THIRD YEAR

3rd year – 1st Semester

Courses	No. of Hours		Units	Prerequisites
	Lec	Lab/Field/Drafting		
Logic Circuits and Design	3	3	4	Fundamentals of Electronic Circuits
Operating Systems	3	0	3	Data Structures and Algorithms
Data and Digital Communications	3	0	3	Fundamentals of Electronic Circuits
Introduction to HDL	0	3	1	Programming Logic and Design; Fundamentals of Electronic Circuits
Feedback and Control Systems	3	0	3	Numerical Methods; Fundamentals of Electrical Circuits
Fundamentals of Mixed Signals and Sensors	3	0	3	Fundamentals of Electronic Circuits
Computer Engineering Drafting and Design	0	3	1	Fundamentals of Electronic Circuits
Cognate / Elective Course 1**			3	3 rd Year Standing*
TOTAL	15	9	21	

3rd year – 2nd semester

Courses	No. of Hours		Units	Prerequisites
	Lec	Lab/Field/Drafting		
Basic Occupational Health and Safety	3	0	3	3 rd Year Standing*
Computer Networks and Security	3	3	4	Data and Digital Communications
Microprocessors	3	3	4	Logic Circuits and Design
Methods of Research	2	0	2	Engineering Data Analysis; Purposive Communication; Logic Circuits and Design
Technopreneurship	3	0	3	3 rd Year Standing*
Ethics	3	0	3	
CpE Laws and Professional Practice	2	0	2	3 rd Year Standing*
Cognate/Elective Course 2**			3	Cognate/Track Course 1
TOTAL	19	6	24	

FOURTH YEAR

4th year – 1st semester

Courses	No. of Hours		Units	Prerequisites
	Lec	Lab/Field/Drafting		
Embedded Systems	3	3	4	Microprocessors
Computer Architecture and Organization	3	3	4	Microprocessors
Emerging Technologies in CpE	3	0	3	4 th Year Standing*
CpE Practice and Design 1	0	3	1	Microprocessors; Methods of Research
Digital Signal Processing	3	3	4	Feedback and Control Systems
GEC Elective 2	3	0	3	
Cognate/Elective Course 3**			3	Cognate/Track Course 2
TOTAL	15	12	22	



4th year – 2nd semester

Courses	No. of Hours		Units	Prerequisites
	Lec	Lab/Field/Drafting		
CpE Practice and Design 2	0	6	2	CpE Practice and Design 1
Seminars and Fieldtrips	0	3	1	4 th Year Standing*
On the Job Training	3	240***	3	4th Year Standing*
GEC Elective 3	3	0	3	
TOTAL	6	246	9	

Suggested Cognates/Electives

(The program has an option to include additional cognates/electives.)

Embedded Systems

Courses	No. of Hours		No. of Hours
	Lab	Lab/Field/Drafting	
Embedded Systems 1			3
Embedded Systems 2			3
Embedded Systems 3			3

Microelectronics

Courses	No. of Hours		No. of Hours
	Lab	Lab/Field/Drafting	
Microelectronics 1			3
Microelectronics 2			3
Microelectronics 3			3

Software Development

Courses	No. of Hours		No. of Hours
	Lab	Lab/Field/Drafting	
Software Development 1			3
Software Development 2			3
Software Development 3			3

System and Network Administration

Courses	No. of Hours		No. of Hours
	Lab	Lab/Field/Drafting	
System and Network Administration 1			3
System and Network Administration 2			3
System and Network Administration 3			3



Technopreneurship

Courses	No. of Hours		No. of Hours
	Lab	Lab/Field/Drafting	
Technopreneurship 1			3
Technopreneurship 2			3
Technopreneurship 3			3

- * The n^{th} year standing means that the student must have completed at least 75% of the load requirements of the previous year level.
- ** The courses in track specializations should be related.
- *** 80 hours per unit of field work.

Section 12. Sample Curriculum Map

Refer to **Annex II** for the Minimum Program Outcomes and a Sample Curriculum Map. The HEI may develop own Curriculum Map.

Section 13. Description of Outcomes-Based Teaching and Learning

Outcomes-based teaching and learning (OBTL) is an approach where teaching and learning activities are developed to support the learning outcomes (University of Hong Kong, 2007). It is a student-centered approach for the delivery of educational programs where the curriculum topics in a program and the courses contained in it are expressed as the intended outcomes for students to learn. It is an approach in which teachers facilitate and students find themselves actively engaged in their learning.

Its primary focus is the clear statement of what students should be able to do after taking a course, known as the Intended Learning Outcomes (ILOs). The ILOs describe what the learners will be able to do when they have completed their course or program. These are statements, written from the students' perspective, indicating the level of understanding and performance they are expected to achieve as a result of engaging in teaching and learning experience (Biggs and Tang, 2007). Once the ILOs have been determined, the next step in OBTL is to design the Teaching / Learning Activities (TLAs) which require students to actively participate in the construction of their new knowledge and abilities. A TLA is any activity which stimulates, encourages or facilitates learning of one or more intended learning outcome. The final OBTL component is the Assessment Tasks (ATs), which measure how well students can use their new abilities to solve real-world problems, design, demonstrate creativity, and communicate effectively, among others. An AT can be any method of assessing how well a set of ILO has been achieved.

A key component of a course design using OBTL is the constructive alignment of ILOs, TLAs, and ATs. This design methodology requires the Intended Learning Outcomes to be developed first, and then the Teaching / Learning Activities and Assessment Tasks are developed based on the ILOs. (Biggs, 1999)



“Constructive” refers to the idea that students construct meaning through relevant learning activities; “alignment” refers to the situation when teaching and learning activities, and assessment tasks, are aligned to the Intended Learning Outcomes by using the verbs stipulated in the ILOs. Constructive alignment provides the “how-to” by stating that the TLAs and the assessment tasks activate the same verbs as in the ILOs. (Biggs and Tang, 1999)

The OBTL approach shall be reflected in the Course Syllabus to be implemented by the faculty.

Section 14. Course Syllabus and Course Specifications

The Course Syllabus must contain at least the following components:

- 14.1. General Course Information (Title, Description, Code, Credit Units, Prerequisites)
- 14.2 Links to Program Outcomes
- 14.3 Course Outcomes
- 14.4 Course Outline (Including Unit Outcomes)
- 14.5 Teaching and Learning Activities
- 14.6 Assessment Methods
- 14.7 Final Grade Evaluation
- 14.8 Learning Resources
- 14.9 Course Policies and Standards
- 14.10 Effectivity and Revision Information

See Annex III for Sample Course Specifications for the courses listed in the suggested Curriculum Map.

ARTICLE VI REQUIRED RESOURCES

This article covers the specific required resources for the BS Computer Engineering program.

All other requirements on Administration, Library and Laboratory facilities, and buildings for the BS Engineering Program are contained in CMO No. 86, s. 2017, Policies, Standards and Guidelines for Requirements Common to all BS Engineering and Bachelor of Engineering Technology Programs issued by the Commission.

Section 15. Administration

The administration of the college of engineering must provide academic governance and leadership to engineering programs by exerting efforts to achieve program educational objectives and program outcomes. As such, the college must have a full-time dean and full-time department or program chair who are adept in the principles of outcomes-based



education and are trained to implement the elements of OBE and OBTL required by CMO 37 s2012.

There shall be a full-time Department/Program Chair/Coordinator who will lead the program in curriculum planning, implementation, monitoring, review, and evaluation of BSCpE program. The College Dean may serve as concurrent Department or Program Chair when appropriate.

The qualifications of the Department/Program Chair/Coordinator of BSCpE program:

- a) Shall be a Professional Computer Engineer, if applicable;
- b) Shall be holder of any of the following Master's degree
 - (1) Master of Science in Computer Engineering
 - (2) Master of Engineering in Computer Engineering
 - (3) Master of Engineering Education in Computer Engineering
 - (4) Master of Engineering Program Major in Computer Engineering
 - (5) Master of Science in Engineering Major in Computer Engineering; and
- c) Shall have a minimum teaching experience of not less than three (3) years preferably with industry practice

The Department/Program Chair to carry out his/her administrative function must be given a teaching load of not more than 50% of regular teaching load.

Section 16. Faculty

16.1 Requirements

There shall be adequate number of competent and qualified faculty to teach professional courses of BSCpE program and appropriate student-faculty ratio to effectively implement the minimum curricular requirements. The program shall not be dependent on single faculty handling professional courses.

In addition, by AY 2018-2019, thirty-five percent (35%) of the total full-time faculty members teaching professional courses in BSCpE must be holder of Master's degree in CpE or allied programs and preferably Doctoral degree in CpE or allied programs. Faculty members teaching professional courses must be a Certified Computer Engineer, if applicable.

All other full-time faculty of the program, including those teaching Mathematics, Sciences, Computing, and General Education courses, must also possess at least Master's degrees relevant to their courses being taught and research specializations by AY 2018-2019.

Faculty members teaching professional courses that require industry certification shall have valid industry certification.



Faculty members teaching CpE Design preferably shall have relevant industry immersion or experience.

All faculty members must undergo training in the principles of OBE and the practice of OBTL using various modes of teaching and learning activities and appropriate outcomes-based assessment.

16.2 Duties

The faculty shall be actively involved in the following areas of implementation of CpE program:

- (1) curriculum review, decision-making, and implementation of the academic program
- (2) program assessment and evaluation, and implementation of continuous improvement of the program
- (3) development, improvement, and achievement of course outcomes (COs)
- (4) enrichment of teaching and learning activities (TLAs)
- (5) development and improvement of assessment tasks, constructively aligned with COs and TLAs
- (6) student advising activities of the program
- (7) research and scholarly work
- (8) professional services offered by the program
- (9) linkage and extension work

Section 17. Library and other Learning Resources

The library services and other learning resources are covered by CMO No. 86, s. 2017, Policies, Standards and Guidelines for Requirements Common to all BS Engineering and Bachelor of Engineering Technology Programs.

Section 18. Laboratory Equipment and Resources

18.1 Facilities

Facilities are covered by CMO No. 86, s. 2017, Policies, Standards and Guidelines for Requirements Common to all BS Engineering Programs

18.2 Laboratories for the BSCpE Program

1. Chemistry for Engineers
2. Physics for Engineers
3. Fundamentals of Electrical Circuits
4. Fundamentals of Electronic Circuits
5. Microprocessors
6. Logic Circuits and Design
7. Computer Architecture and Organization
8. Computer Networks and Security
9. Embedded Systems



10. Computer Engineering Drafting and Design
11. Programming Logic and Design
12. Data Structures and Algorithms
13. Object Oriented Programming
14. Software Design
15. Digital Signal Processing
16. Introduction to HDL

18.3 Modernization of Facilities

Each school/college of engineering shall have a program for the continuing modernization and upgrading of its instructional laboratories, facilities, and equipment. The said program shall have an adequate annual allocation in accordance with the financial capability of the school.

18.4 Calibration of Equipment

Each school/college of engineering shall ensure that the measuring instruments in its laboratories are recalibrated regularly. The date of the last calibration of the measuring instrument shall be indicated on each instrument.

ARTICLE VII COMPLIANCE OF HEIs

Section 19 Full Compliance with CMO 37, s. 2012

Before the start of AY 2018-2019, all HEIs offering BS in Computer Engineering programs must show evidence of full compliance with CMO 37, s. 2012 (Establishment of an Outcomes-Based Education System) by the following actions:

19.1 CMO 37 Monitoring Workbook and Self-Assessment Rubric

The Commission, through its Regional offices or the TPET Website shall make available to all HEIs currently offering or applying to offer BS Computer Engineering programs a Monitoring Workbook (CMO 37-MW-2017-HEI-BSCpE) and Self-Assessment Rubric (SAR) (CMO-37-HEI-SAR-2017-BSCpE).

The five-year BCpE curriculum shall be the basis of the monitoring. The completed Monitoring Workbook with a List of Supporting Evidences and Self-Assessment Rubric must be submitted to CHED or online through the CHED TPET website (www.ched-tpet.org) within 30 working days after the effectivity of this CMO. Failure to submit these documents will disqualify the concerned HEIs from continuing or starting their BSAeE programs in AY 2018-2019.

19.2 Review of Submitted Forms by CHED

CHED shall review the submitted Monitoring Workbooks and Self-Assessment Rubrics, and may schedule monitoring visits to the HEI thereafter. These visits shall determine the extent of compliance of the



concerned HEI with CMO 37, s. 2012. HEIs with BCpE programs with low SAR total scores may be asked to submit a one- or two-year development plan to CHED.

19.3 Exemptions

HEIs with BSCpE programs that have applied as COEs/CODs during AY 2015-2016 and whose applications have been approved as COE or COD shall not be required to comply with Section 19.1 and 19.2. Instead, these HEIs must submit only their proposed four-year curriculum, corresponding curriculum map, and program of study using the Application Workbook for AY 2018-2019 (AW-2018-HEI-BSCpE). See Section 20. Those HEIs whose COD/COE applications were disapproved for AY 2018-2019 must still comply with Sections 19.1 and 19.2.

Section 20 Application Workbook for AY 2018-2019

HEIs currently offering the BSCpE program for AY 2018-2019 shall be made to complete a new Application Workbook (AW-2018-HEI-BSCpE) which shall be made available through CHED or downloadable from the CHED-TPET website. The Application Workbook shall be completed and submitted to CHED or uploaded to the CHED-TPET website before the start of AY 2018-2019.

Section 21 Approval of Application

All HEIs with BSCpE programs with COE or COD status submitting their completed Application Workbooks shall automatically receive certifications from CHED and shall be given approval to implement their programs beginning AY 2018-2019.

Other concerned HEIs which have submitted their CMO Monitoring Workbooks, Self-Assessment Rubrics, and Application Workbook shall be given conditional approval by CHED to start offering their new BSIE Curriculum following this CMO effective AY 2018-2019. CHED shall, however, conduct monitoring of HEIs to assure complete compliance of this PSG within the transitory period, during which HEIs with BSCpE programs with weak implementation may be asked to submit developmental plans, which shall be subject to constant monitoring.

ARTICLE VIII TRANSITORY, REPEALING and EFFECTIVITY PROVISIONS

Section 22 Transitory Provision

All private HEIs, state universities and colleges, and local universities and colleges with existing authorization to operate the Bachelor of Science in Computer Engineering program are hereby given a period of three (3) years from the effectivity thereof to fully comply with all the



requirements in this CMO. However, the prescribed minimum curricular requirements in this CMO shall be implemented starting AY 2018-2019.

Section 23 Repealing Clause

Any provision of this Order, which may thereafter be held invalid, shall not affect the remaining provisions.

All CHED issuances or part thereof inconsistent with the provision in this CMO shall be deemed modified or repealed.

Section 24 Effectivity Clause

This CMO shall take effect fifteen (15) days after its publication in the Official Gazette or in a newspaper of general circulation. This CMO shall be implemented beginning AY 2018-2019.

Quezon City, Philippines December 4, 2017

For the Commission:


PATRICIA B. LICUANAN, Ph.D.
Chairperson

Attachments:

- Annex I – Competency Standards for an Industrial Engineer
- Annex II – Minimum Program Outcomes and Sample Curriculum Map
- Annex III – Sample Course Specifications
- Annex IV – Laboratory Requirements
 - A. Natural/Physical Sciences
 - B. Professional Courses
- Annex V – Sample Course Syllabus

